

- Description

This module manages a smart toll booth traffic monitoring system, ensuring that:

1. **The number of vehicles passing through the toll station is continuously monitored.**
2. **If traffic congestion exceeds a predefined threshold, additional toll lanes are opened to improve flow.**
3. **If traffic volume drops below a minimum threshold, some lanes are closed to optimize resource usage.**
4. **A log of all vehicle counts, congestion alerts, and lane adjustments is maintained for operational efficiency.**

This system helps prevent toll station bottlenecks, optimize lane usage, and improve overall traffic efficiency.

- Specification

module SmartTollBoothMonitoring

var currentTraffic, maxTraffic, minTraffic, openLanes: integer

var log: sequence of string

process vehicleArrives(numVehicles: integer)

functional scenario form:

currentTraffic <= maxTraffic && log = log $\oplus$ ("Low Traffic: Closing Unused Lanes."), Vehicles entered, lanes adjusted.

||

currentTraffic > maxTraffic && log = log $\oplus$ ("Heavy Congestion Alert: Additional Lanes Opened."), Heavy congestion warning triggered, opening additional lanes.

end process

process vehicleExits(numVehicles: integer)

functional scenario form:

log = log $\oplus$ ("Exit: " + numVehicles + " vehicles | New Traffic: " + currentTraffic + " | Open Lanes: " + openLanes), Vehicles exited, lanes adjusted.

end process

end module

- Java Code

```
package code;
import java.util.*;
class TollBooth {
    private int maxTraffic;
    private int minTraffic;
    private int currentTraffic;
    private int openLanes;
    private List<String> log;
```

```

    public TollBooth(int maxTraffic, int minTraffic, int initialTraffic
) {
    this.maxTraffic = maxTraffic;
    this.minTraffic = minTraffic;
    this.currentTraffic = initialTraffic;
    this.openLanes = 5;
    this.log = new ArrayList<>();
}
    public void vehicleArrives(int numVehicles) {
        currentTraffic += numVehicles;
        adjustLanes();
        String message = String.format("Vehicles Arrived: %d | Current
Traffic: %d | Open Lanes: %d", numVehicles, currentTraffic, openLanes);
        if (currentTraffic > maxTraffic) {
            message += " | WARNING: Heavy Congestion! Opening Additiona
l Lanes.";
        } else if (currentTraffic < minTraffic) {
            message += " | Traffic Low. Closing Unused Lanes.";
        }
        System.out.println(message);
        log.add(message);
    }
    public void vehicleExits(int numVehicles) {
        currentTraffic = Math.max(0, currentTraffic - numVehicles);
        adjustLanes();
        System.out.printf("Vehicles Exited: %d | Current Traffic: %d |
Open Lanes: %d\n", numVehicles, currentTraffic, openLanes);
        log.add(String.format("Exit: %d vehicles | New Traffic: %d | Op
en Lanes: %d", numVehicles, currentTraffic, openLanes));
    }
    private void adjustLanes() {
        if (currentTraffic > maxTraffic * 0.8) {
            openLanes = 8;
        } else if (currentTraffic < minTraffic) {
            openLanes = 3;
        } else {
            openLanes = 5;
        }
    }
}
    public void printLog() {
        System.out.println("\nToll Booth Traffic Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

```

```

    }
}
public class c44_SmartTollBoothSystem {
    public static void main(String[] args) {
        TollBooth tollBooth = new TollBooth(100, 30, 50);
        // testcase-VT:
        // tollBooth.vehicleArrives(40);
        // tollBooth.vehicleArrives(20);
        // tollBooth.vehicleExits(30);
        // tollBooth.vehicleArrives(10);
        // tollBooth.vehicleExits(50);
        // tollBooth.vehicleArrives(-188);
        // tollBooth.vehicleArrives(-267);
        // tollBooth.vehicleArrives(-456);
        // tollBooth.vehicleArrives(-117);
        // tollBooth.vehicleArrives(426);
        // tollBooth.vehicleArrives(-459);
        // tollBooth.vehicleArrives(107);
        // tollBooth.vehicleArrives(-50);
        // tollBooth.vehicleArrives(-317);
        // tollBooth.vehicleArrives(-53);
        // tollBooth.vehicleArrives(70);
        // tollBooth.vehicleArrives(362);
        // tollBooth.vehicleArrives(364);
        // tollBooth.vehicleArrives(-107);
        // tollBooth.vehicleArrives(-35);
        // tollBooth.vehicleArrives(-178);
        // tollBooth.vehicleArrives(409);
        // tollBooth.vehicleArrives(-166);
        // tollBooth.vehicleArrives(133);
        // tollBooth.vehicleArrives(-130);
        // tollBooth.vehicleArrives(-153);
        // tollBooth.vehicleArrives(-205);
        // tollBooth.vehicleArrives(-404);
        // tollBooth.vehicleArrives(-325);
        // tollBooth.vehicleArrives(225);
        // tollBooth.vehicleArrives(31);
        // tollBooth.vehicleArrives(317);
        // tollBooth.vehicleArrives(-141);
        // tollBooth.vehicleArrives(223);
        // tollBooth.vehicleArrives(-411);
        // tollBooth.vehicleArrives(-242);
        // tollBooth.vehicleArrives(-448);
        // tollBooth.vehicleArrives(159);
    }
}

```

```
// tollBooth.vehicleArrives(-271);
// tollBooth.vehicleArrives(-373);
// tollBooth.vehicleArrives(482);
// tollBooth.vehicleArrives(99);
// tollBooth.vehicleArrives(445);
// tollBooth.vehicleArrives(-426);
// tollBooth.vehicleArrives(-272);
// tollBooth.vehicleArrives(153);
// tollBooth.vehicleArrives(-90);
// tollBooth.vehicleArrives(-156);
// tollBooth.vehicleArrives(15);
// tollBooth.vehicleArrives(-42);
// tollBooth.vehicleExits(-188);
// tollBooth.vehicleExits(-240);
// tollBooth.vehicleExits(400);
// tollBooth.vehicleExits(-148);
// tollBooth.vehicleExits(-181);
// tollBooth.vehicleExits(-78);
// tollBooth.vehicleExits(-227);
// tollBooth.vehicleExits(-257);
// tollBooth.vehicleExits(-480);
// tollBooth.vehicleExits(102);
// tollBooth.vehicleExits(-200);
// tollBooth.vehicleExits(310);
// tollBooth.vehicleExits(454);
// tollBooth.vehicleExits(210);
// tollBooth.vehicleExits(363);
// tollBooth.vehicleExits(382);
// tollBooth.vehicleExits(-481);
// tollBooth.vehicleExits(-290);
// tollBooth.vehicleExits(73);
// tollBooth.vehicleExits(466);
// tollBooth.vehicleExits(486);
// tollBooth.vehicleExits(-181);
// tollBooth.vehicleExits(25);
// tollBooth.vehicleExits(-314);
// tollBooth.vehicleExits(292);
// tollBooth.vehicleExits(-307);
// tollBooth.vehicleExits(-191);
// tollBooth.vehicleExits(-18);
// tollBooth.vehicleExits(381);
// tollBooth.vehicleExits(-77);
// tollBooth.vehicleExits(145);
// tollBooth.vehicleExits(377);
```

```
// tollBooth.vehicleExits(39);
// tollBooth.vehicleExits(-382);
// tollBooth.vehicleExits(-409);
// tollBooth.vehicleExits(440);
// tollBooth.vehicleExits(448);
// tollBooth.vehicleExits(-104);
// tollBooth.vehicleExits(8);
// tollBooth.vehicleExits(261);
// tollBooth.vehicleExits(-456);
// tollBooth.vehicleExits(-44);
// tollBooth.vehicleExits(-152);
// tollBooth.vehicleExits(-137);
// tollBooth.vehicleExits(-58);
// tollBooth.vehicleExits(335);
// tollBooth.vehicleExits(-159);
// tollBooth.vehicleExits(-153);
// tollBooth.vehicleExits(-16);
// tollBooth.vehicleExits(-325);
// tesecase-FT:
// tollBooth.vehicleArrives(93);
// tollBooth.vehicleArrives(1629);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(394811);
// tollBooth.vehicleArrives(0);
// tollBooth.vehicleArrives(0);
// tollBooth.vehicleArrives(0);
// tollBooth.vehicleArrives(0);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(445446);
// tollBooth.vehicleArrives(1494022);
// tollBooth.vehicleArrives(52230);
// tollBooth.vehicleArrives(13370928);
// tollBooth.vehicleArrives(100663296);
// tollBooth.vehicleArrives(100663296);
// tollBooth.vehicleArrives(100663296);
// tollBooth.vehicleArrives(100663296);
// tollBooth.vehicleArrives(100674560);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(-892679478);
// tollBooth.vehicleArrives(-892679478);
// tollBooth.vehicleArrives(1739);
```

```
// tollBooth.vehicleArrives(394955);
// tollBooth.vehicleArrives(1739);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(62982);
// tollBooth.vehicleArrives(0);
// tollBooth.vehicleArrives(0);
// tollBooth.vehicleArrives(0);
// tollBooth.vehicleArrives(255);
// tollBooth.vehicleArrives(255);
// tollBooth.vehicleArrives(47);
// tollBooth.vehicleArrives(12038);
// tollBooth.vehicleArrives(12079);
// tollBooth.vehicleArrives(1);
// tollBooth.vehicleArrives(5);
// tollBooth.vehicleArrives(9477);
// tollBooth.vehicleArrives(9478);
// tollBooth.vehicleArrives(2426406);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(394758);
// tollBooth.vehicleArrives(402694);
// tollBooth.vehicleArrives(103089670);
// tollBooth.vehicleArrives(70);
// tollBooth.vehicleArrives(17926);
// tollBooth.vehicleArrives(4589062);
// tollBooth.vehicleArrives(4589062);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(1733);
// tollBooth.vehicleArrives(1597);
// tollBooth.vehicleArrives(104723957);
// tollBooth.vehicleArrives(104723957);
// tollBooth.vehicleArrives(117440511);
// tollBooth.vehicleArrives(47);
// tollBooth.vehicleArrives(59);
// tollBooth.vehicleArrives(64);
// tollBooth.vehicleArrives(15168);
// tollBooth.vehicleArrives(15296);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(11732486);
// tollBooth.vehicleArrives(45830);
// tollBooth.vehicleArrives(-1291845632);
```

```
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(14);
// tollBooth.vehicleArrives(14);
// tollBooth.vehicleArrives(234881024);
// tollBooth.vehicleArrives(234881024);
// tollBooth.vehicleArrives(117440511);
// tollBooth.vehicleArrives(117440511);
// tollBooth.vehicleArrives(108724223);
// tollBooth.vehicleArrives(108724223);
// tollBooth.vehicleArrives(-250);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(125);
// tollBooth.vehicleArrives(104);
// tollBooth.vehicleArrives(360);
// tollBooth.vehicleArrives(9320);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(518);
// tollBooth.vehicleArrives(1542);
// tollBooth.vehicleArrives(63994);
// tollBooth.vehicleArrives(64249);
// tollBooth.vehicleArrives(393217);
// tollBooth.vehicleArrives(393222);
// tollBooth.vehicleArrives(393216);
// tollBooth.vehicleArrives(2490368);
// tollBooth.vehicleArrives(2490369);
// tollBooth.vehicleArrives(6);
// tollBooth.vehicleArrives(22);
// tollBooth.vehicleArrives(825307441);
// tollBooth.vehicleArrives(70);
// tollBooth.vehicleArrives(71);
// tollBooth.vehicleArrives(1607);
// testcase-Z3:
// tollBooth.vehicleArrives(40);
// tollBooth.vehicleArrives(20);
// tollBooth.vehicleExits(30);
// tollBooth.vehicleArrives(10);
// tollBooth.vehicleExits(50);
// tollBooth.vehicleArrives(-188);
// tollBooth.vehicleArrives(-267);
// tollBooth.vehicleArrives(-456);
// tollBooth.vehicleArrives(-117);
// tollBooth.vehicleArrives(426);
// tollBooth.vehicleArrives(-459);
// tollBooth.vehicleArrives(107);
```

```
// tollBooth.vehicleArrives(-50);
// tollBooth.vehicleArrives(-317);
// tollBooth.vehicleArrives(-53);
// tollBooth.vehicleArrives(70);
// tollBooth.vehicleArrives(362);
// tollBooth.vehicleArrives(364);
// tollBooth.vehicleArrives(-107);
// tollBooth.vehicleArrives(-35);
// tollBooth.vehicleArrives(-178);
// tollBooth.vehicleArrives(409);
// tollBooth.vehicleArrives(-166);
// tollBooth.vehicleArrives(133);
// tollBooth.vehicleArrives(-130);
// tollBooth.vehicleArrives(-153);
// tollBooth.vehicleArrives(-205);
// tollBooth.vehicleArrives(-404);
// tollBooth.vehicleArrives(-325);
// tollBooth.vehicleArrives(225);
// tollBooth.vehicleArrives(31);
// tollBooth.vehicleArrives(317);
// tollBooth.vehicleArrives(-141);
// tollBooth.vehicleArrives(223);
// tollBooth.vehicleArrives(-411);
// tollBooth.vehicleArrives(-242);
// tollBooth.vehicleArrives(-448);
// tollBooth.vehicleArrives(159);
// tollBooth.vehicleArrives(-271);
// tollBooth.vehicleArrives(-373);
// tollBooth.vehicleArrives(482);
// tollBooth.vehicleArrives(99);
// tollBooth.vehicleArrives(445);
// tollBooth.vehicleArrives(-426);
// tollBooth.vehicleArrives(-272);
// tollBooth.vehicleArrives(153);
// tollBooth.vehicleArrives(-90);
// tollBooth.vehicleArrives(-156);
// tollBooth.vehicleArrives(15);
// tollBooth.vehicleArrives(-42);
// tollBooth.vehicleExits(-188);
// tollBooth.vehicleExits(-240);
// tollBooth.vehicleExits(400);
// tollBooth.vehicleExits(-148);
// tollBooth.vehicleExits(-181);
// tollBooth.vehicleExits(-78);
```

```
// tollBooth.vehicleExits(-227);
// tollBooth.vehicleExits(-257);
// tollBooth.vehicleExits(-480);
// tollBooth.vehicleExits(102);
// tollBooth.vehicleExits(-200);
// tollBooth.vehicleExits(310);
// tollBooth.vehicleExits(454);
// tollBooth.vehicleExits(210);
// tollBooth.vehicleExits(363);
// tollBooth.vehicleExits(382);
// tollBooth.vehicleExits(-481);
// tollBooth.vehicleExits(-290);
// tollBooth.vehicleExits(73);
// tollBooth.vehicleExits(466);
// tollBooth.vehicleExits(486);
// tollBooth.vehicleExits(-181);
// tollBooth.vehicleExits(25);
// tollBooth.vehicleExits(-314);
// tollBooth.vehicleExits(292);
// tollBooth.vehicleExits(-307);
// tollBooth.vehicleExits(-191);
// tollBooth.vehicleExits(-18);
// tollBooth.vehicleExits(381);
// tollBooth.vehicleExits(-77);
// tollBooth.vehicleExits(145);
// tollBooth.vehicleExits(377);
// tollBooth.vehicleExits(39);
// tollBooth.vehicleExits(-382);
// tollBooth.vehicleExits(-409);
// tollBooth.vehicleExits(440);
// tollBooth.vehicleExits(448);
// tollBooth.vehicleExits(-104);
// tollBooth.vehicleExits(8);
// tollBooth.vehicleExits(261);
// tollBooth.vehicleExits(-456);
// tollBooth.vehicleExits(-44);
// tollBooth.vehicleExits(-152);
// tollBooth.vehicleExits(-137);
// tollBooth.vehicleExits(-58);
// tollBooth.vehicleExits(335);
// tollBooth.vehicleExits(-159);
// tollBooth.vehicleExits(-153);
// tollBooth.vehicleExits(-16);
// tollBooth.vehicleExits(-325);
```

```
// testcase-UVT:
tollBooth.vehicleArrives(40);
tollBooth.vehicleArrives(20);
tollBooth.vehicleExits(30);
tollBooth.vehicleArrives(10);
tollBooth.vehicleExits(50);
tollBooth.vehicleArrives(-188);
tollBooth.vehicleArrives(-267);
tollBooth.vehicleArrives(-456);
tollBooth.vehicleArrives(-117);
tollBooth.vehicleArrives(426);
tollBooth.vehicleArrives(-459);
tollBooth.vehicleArrives(107);
tollBooth.vehicleArrives(-50);
tollBooth.vehicleArrives(-317);
tollBooth.vehicleArrives(-53);
tollBooth.vehicleArrives(70);
tollBooth.vehicleArrives(362);
tollBooth.vehicleArrives(364);
tollBooth.vehicleArrives(-107);
tollBooth.vehicleArrives(-35);
tollBooth.vehicleArrives(-178);
tollBooth.vehicleArrives(409);
tollBooth.vehicleArrives(-166);
tollBooth.vehicleArrives(133);
tollBooth.vehicleArrives(-130);
tollBooth.vehicleArrives(-153);
tollBooth.vehicleArrives(-205);
tollBooth.vehicleArrives(-404);
tollBooth.vehicleArrives(-325);
tollBooth.vehicleArrives(225);
tollBooth.vehicleArrives(31);
tollBooth.vehicleArrives(317);
tollBooth.vehicleArrives(-141);
tollBooth.vehicleArrives(223);
tollBooth.vehicleArrives(-411);
tollBooth.vehicleArrives(-242);
tollBooth.vehicleArrives(-448);
tollBooth.vehicleArrives(159);
tollBooth.vehicleArrives(-271);
tollBooth.vehicleArrives(-373);
tollBooth.vehicleArrives(482);
tollBooth.vehicleArrives(99);
```

```
tollBooth.vehicleArrives(445);
tollBooth.vehicleArrives(-426);
tollBooth.vehicleArrives(-272);
tollBooth.vehicleArrives(153);
tollBooth.vehicleArrives(-90);
tollBooth.vehicleArrives(-156);
tollBooth.vehicleArrives(15);
tollBooth.vehicleArrives(-42);
tollBooth.vehicleExits(-188);
tollBooth.vehicleExits(-240);
tollBooth.vehicleExits(400);
tollBooth.vehicleExits(-148);
tollBooth.vehicleExits(-181);
tollBooth.vehicleExits(-78);
tollBooth.vehicleExits(-227);
tollBooth.vehicleExits(-257);
tollBooth.vehicleExits(-480);
tollBooth.vehicleExits(102);
tollBooth.vehicleExits(-200);
tollBooth.vehicleExits(310);
tollBooth.vehicleExits(454);
tollBooth.vehicleExits(210);
tollBooth.vehicleExits(363);
tollBooth.vehicleExits(382);
tollBooth.vehicleExits(-481);
tollBooth.vehicleExits(-290);
tollBooth.vehicleExits(73);
tollBooth.vehicleExits(466);
tollBooth.vehicleExits(486);
tollBooth.vehicleExits(-181);
tollBooth.vehicleExits(25);
tollBooth.vehicleExits(-314);
tollBooth.vehicleExits(292);
tollBooth.vehicleExits(-307);
tollBooth.vehicleExits(-191);
tollBooth.vehicleExits(-18);
tollBooth.vehicleExits(381);
tollBooth.vehicleExits(-77);
tollBooth.vehicleExits(145);
tollBooth.vehicleExits(377);
tollBooth.vehicleExits(39);
tollBooth.vehicleExits(-382);
tollBooth.vehicleExits(-409);
tollBooth.vehicleExits(440);
```

```

        tollBooth.vehicleExits(448);
        tollBooth.vehicleExits(-104);
        tollBooth.vehicleExits(8);
        tollBooth.vehicleExits(261);
        tollBooth.vehicleExits(-456);
        tollBooth.vehicleExits(-44);
        tollBooth.vehicleExits(-152);
        tollBooth.vehicleExits(-137);
        tollBooth.vehicleExits(-58);
        tollBooth.vehicleExits(335);
        tollBooth.vehicleExits(-159);
        tollBooth.vehicleExits(-153);
        tollBooth.vehicleExits(-16);
        tollBooth.vehicleExits(-325);

//      tollBooth.printLog();
    }
}

```

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c44_SmartTollBoothSystem.java	80.0 %	8	2	10
> TollBooth	80.0 %	8	2	10
> c44_SmartTollBoothSystem		0	0	0

3. Vibration testing instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.2 %	470	20,858	21,328
▼ src	2.2 %	470	20,858	21,328
▼ code	2.2 %	470	20,858	21,328
▼ c44_SmartTollBoothSystem.java	95.5 %	470	22	492
> TollBooth	89.5 %	162	19	181
> c44_SmartTollBoothSystem	99.0 %	308	3	311

4. Fuzz testing branch coverage: 60%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c44_SmartTollBoothSystem.java	60.0 %	6	4	10
> c44_SmartTollBoothSystem		0	0	0
> TollBooth	60.0 %	6	4	10

5. Fuzz testing instruction coverage: 82.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.9 %	408	20,932	21,340
▼ src	1.9 %	408	20,932	21,340
▼ code	1.9 %	408	20,932	21,340
▼ c44_SmartTollBoothSystem.java	82.9 %	408	84	492
> c44_SmartTollBoothSystem	99.0 %	308	3	311
> TollBooth	55.2 %	100	81	181

6. Z3 branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	832	840
▼ src	1.0 %	8	832	840
▼ code	1.0 %	8	832	840
▼ c44_SmartTollBoothSystem.java	80.0 %	8	2	10
> TollBooth	80.0 %	8	2	10
> c44_SmartTollBoothSystem		0	0	0

7. Z3 instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.1 %	470	22,206	22,676
▼ src	2.1 %	470	22,206	22,676
▼ code	2.1 %	470	22,206	22,676
▼ c44_SmartTollBoothSystem.java	95.5 %	470	22	492
> TollBooth	89.5 %	162	19	181
> c44_SmartTollBoothSystem	99.0 %	308	3	311

8. Unimproved V-method branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	832	840
▼ src	1.0 %	8	832	840
▼ code	1.0 %	8	832	840
▼ c44_SmartTollBoothSystem.java	80.0 %	8	2	10
> TollBooth	80.0 %	8	2	10
> c44_SmartTollBoothSystem		0	0	0

9. Unimproved V-method instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.1 %	470	22,206	22,676
▼ src	2.1 %	470	22,206	22,676
▼ code	2.1 %	470	22,206	22,676
▼ c44_SmartTollBoothSystem.java	95.5 %	470	22	492
> TollBooth	89.5 %	162	19	181
> c44_SmartTollBoothSystem	99.0 %	308	3	311